Abstract

Leon Kaushik Deka · Third Semester AIML Mini Project

CCTV cameras are present in many different locations including hostels, banks, shops etc. However often these cameras capture more information than required. For example a camera in a hostel corridor may capture personal clothing of residents or a camera above the billing counter may capture customers entering their UPI PIN. This video can be accessed by anyone with access to the control room or if someone were to hack into the system.

This project aims to create an AI/ML based CCTV system. It will mask only the sensitive areas of the video. Each frame passes through the AI chip located in the camera, meaning that no unmasked footage can be stored or sent. Using a pose estimation model alongside a pre-trained body part detection model, private areas of the body (nipples, buttocks and genitalia) can be identified and masked. Another pipeline is used to identify phones and POS machines. Hand tracking is also used to identify when someone is entering their UPI PIN as the payee. At this point the screen and hand are masked. The masked areas of the video are permanent. As there is no forensic value to a PIN or private area of the body, the rest of the video can still be used as evidence.

This system has been designed to be used in AI enabled CCTV cameras. A prototype will be created using a laptop and webcam. The prototype will be tested on how well it masks the sensitive areas, how often it masks unnecessary areas and how quickly it can process in real time. This provides a practical solution to CCTV surveillance with a privacy by design approach.